DANISH AHMED

Full-Stack Engineer | AI / LLM Systems Engineer

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Available now | Remote, hybrid or relocation (US, Canada, Europe, UK, UAE) | Open to visa sponsorship

PROFESSIONAL SUMMARY

Full-stack and AI systems engineer who ships production software end to end, from database schema and security model to containerised deployment and a passing test suite. Solo-built and deployed a live e-commerce platform on Next.js 16 / React 19 / TypeScript with a Payload CMS backend, a hardened checkout and eight third-party integrations. Builds multi-agent LLM systems in Python (FastAPI, LangGraph, Celery, MCP), including a portable agent framework of his own design, and has delivered production AI agents for clients in the United States, Latvia and Pakistan. B.S. Computer Science, KIET (coursework completed Fall 2025).

TECHNICAL SKILLS

Languages: TypeScript, JavaScript (ES6+), Python, SQL, C#, HTML5, CSS3, Bash

Frontend: React 19, Next.js 16 (App Router, SSR, SSG, RSC, ISR), Tailwind CSS 4, Zustand, Motion, Material UI, Streamlit, responsive UI

Backend: Node.js, Express, Payload CMS v3, FastAPI, Django, Flask, .NET (C#), REST, GraphQL, Zod, JWT, RBAC, rate limiting, webhooks

Databases: PostgreSQL (RLS, pgvector), MongoDB, Microsoft SQL Server, SQLite, FAISS, schema design, migrations, ETL

AI / ML: LangGraph, LangChain, MCP, RAG, Claude API, OpenAI API, LLaMA / Phi-3, HuggingFace embeddings, Whisper, prompt engineering, scikit-learn, TensorFlow, PyTorch, SHAP, feature engineering, LLM output evaluation

Automation: n8n, Celery (worker and beat), Meta WhatsApp Cloud API, Telegram Bot API, multi-agent orchestration, scheduled and checkpointed pipelines

Cloud / DevOps: Docker, Docker Compose, Hetzner, Contabo, Coolify, Vercel, Railway, Cloudflare R2, Upstash Redis, GitHub Actions CI/CD, Sentry, SSH hardening, UFW, Fail2ban, Tailscale

Testing: Playwright, Vitest, pytest, ESLint, TypeScript strict mode, security auditing, code review

Data: Power BI, DAX, pandas, NumPy, Plotly, Hopsworks feature store, Excel, KPI dashboards

Practices: SEO and JSON-LD structured data, WCAG-AA accessibility practice, web application security, performance optimisation, technical documentation

PROFESSIONAL EXPERIENCE

AI Contractor / Business Analyst - Turing Enterprises, Inc.

United States (Remote, 4-hour PST overlap) | January 2026 - June 2026

- Performed side-by-side evaluations of large language model outputs for quality, accuracy and relevance against established guidelines.
- Fact-checked model-generated content to find inconsistencies, writing structured justifications for every evaluation decision.
- Used browser-based tooling and command-line instructions in evaluation workflows; documented findings for stakeholders.
- Contract engagement of up to 40 hours per week across three successive services agreements.

AI Engineer (Contract) - Sharpen-AI SIA

Riga, Latvia (Remote) | October 2025 - December 2025 | project for a Latvian wood-products e-commerce client

- Built and shipped a production WhatsApp AI agent for an e-commerce client: product inquiries, live stock lookups (Zoho Sheets), automated offer-PDF generation and order routing, with human handoff for payment.
- Designed multi-stage conversational workflows in n8n on the Meta WhatsApp Cloud API, separating simple orders from complex ones with automated human handoff.
- Delivered a handover package: schema definitions, deployment runbook, integration guides and UAT support.

Data Science Intern - 10Pearls (Shine Internship Program)

Karachi, Pakistan (Remote) | September 2025 - November 2025

- Built a serverless air-quality forecasting pipeline: OpenWeather ingestion, 29+ engineered features, daily retraining and rolling 72-hour predictions.
- Benchmarked four models (Random Forest, Gradient Boosting, Ridge, Lasso); best was Ridge Regression, R-squared 0.741 (RMSE 9.14, MAE 7.53).
- Shipped a Streamlit dashboard with live forecasts and SHAP explainability; automated retraining via GitHub Actions on a Hopsworks feature store.

Data Analyst Intern (Aspire Summer Internship), Office of the Provost (IDAR) - Aga Khan University

Karachi, Pakistan (On-site) | July 2024 - August 2024

- Produced user guides and technical documentation (Power Query, DAX and data-model logic) for 9 institutional Power BI dashboards, including the multi-dashboard Teaching & Learning and QTL_net survey workbooks: SCCAF, Teaching Allocation Model, Teaching & Learning, Strategic Plan Monitoring, MBBS Curriculum Mapping, QTL_net Surveys, Alumni Survey 2023, Student Evaluation of Teaching and Pipeline of Student Journey.

- Produced a narrated tutorial video for the Student Evaluation of Teaching dashboard (script, voice-over, slides) and presented the documentation to the Director of IDAR (27 Aug 2024).

Volunteer, Alumni Affairs - Aga Khan University

Karachi, Pakistan (On-site) | September 2022 - June 2023 | 1,127 volunteer hours (AKU HR letter)

- Researched and updated alumni records for AKU's worldwide alumni community (roughly 10,000 at the time) in the Raiser's Edge database and Excel, finding each alumnus's current position and university, matching them to LinkedIn profiles and recording the details and profile links.
- Entered applicants' details into the alumni database and card sheets for alumni-card processing, handling confidential data with administrator access under staff supervision.
- Assisted staff at Alumni Affairs events including PGME graduation, reunions for SONAM, IED and the Medical College (MBBS), the University convocation, the Impact Symposium, internship ceremonies, the Distinguished Alumni Awards and the AKU 40-year programme.

FREELANCE AND CLIENT ENGAGEMENTS

Barakah Abaya - Full-Stack Developer (Freelance)

Karachi, Pakistan | July 2026 - Present | barakahabaya.com

- Sole engineer on a production e-commerce platform (details under Selected Projects).

US travel-guide publisher - Web Developer and Data Engineer (Contract)

United States (Remote) | February 2026 - Present

- Built a WordPress / WooCommerce storefront for the full guide catalogue: AJAX filtering and live search, six retailer buy-links per guide, and a three-state crowdfunding hub the client switches from an admin bar. Live since August 2026.
- Engineered a retailer-link pipeline across 5+ storefronts: scraped 409 Google Play titles with a fuzzy-matching engine, matched 304 PublishDrive titles with an audit log, reconciled ~230 uncatalogued titles and audited 198 IngramSpark titles.
- Built an OCR cover-QA system over ~900 cover images (title-vs-filename mismatch, letterbox and aspect detection, best-source mapping).
- Delivered the OpenClaw AI publishing assistant: all 27 contracted deliverables in 3 days against a 2-week estimate; Phase 2 added 13 features.

Whiteweld & Company / Whiteweld Foundation - Web Developer

United States (Remote) | 2025 | wbbre.com, whiteweldfoundation.org

- Built both live sites (investment firm and child-welfare foundation); development by me, interface design by a designer.

Storybook Homes NJ - Web Developer (Freelance)

United States (Remote) | August 2026 | storybookhomesnj.com

- Built the marketing site for a New Jersey custom home builder: builder profile, project gallery and portfolio listings.

SELECTED PROJECTS

Barakah Abaya - Production E-Commerce Platform

Solo build, live at barakahabaya.com

Next.js 16, React 19, TypeScript, Payload CMS v3, PostgreSQL, Tailwind CSS 4, Zustand, Zod, Docker, Coolify, Cloudflare R2, Upstash Redis, Sentry

- Designed and built the entire stack alone: content model, storefront, checkout, admin panel, blog and a dual taxonomy of categories and curated collections.
- Security-first checkout: server-side price recomputation, stock and blacklist guards, SHA-256-hashed OTP verified in constant time, timing-safe HMAC webhook verification that fails closed, per-endpoint rate limiting and RBAC.
- Fixed an order-confirmation race with an atomic conditional claim and a single DB transaction, so exactly one of OTP verification, payment webhook or redelivery commits stock.
- Ran a security and correctness audit with every finding independently re-verified: 31 issues fixed, 9 false positives rejected (privilege escalation, unprotected writes on eight collections, fail-open webhooks, overselling, discount multi-consumption).
- Test suite: 56 Playwright end-to-end tests and 245 Vitest tests (189 unit, 56 integration); TypeScript and ESLint at zero errors; privilege-escalation attempts return 403 on the deployed app.
- Eight environment-gated integrations (WhatsApp Cloud API, Safepay, PostEx, Resend, Upstash Redis, Cloudflare R2, n8n, Sentry) and seven authored n8n order-lifecycle workflows; SSG with on-demand ISR and JSON-LD structured data.

Agent Core - Portable AI Agent Platform

Personal project, owned outright (source private)

Python, MCP v2, multi-provider LLM routing, SQLite, BM25, pytest

- Provider-agnostic LLM router with retry, cascading fallback and separate data lanes for sensitive and non-sensitive traffic; MCP v2 tool layer on the stable mcp 2.0.0 release.
- Checkpointed unattended execution that resumes after a power cut; per-client deployment with an output sanitiser; scheduled lead monitor across 37 sources.
- Multi-agent crew with BM25 long-term memory; bi-temporal knowledge graph where contradicting facts invalidate rather than delete; deterministic honesty gate, PII redaction gate and refusal gate.

Blink - RAG Policy Assistant with a Local LLM (Final Year Project, among the top five at KIET)

2025 - 2026 | github.com/Danish-Ahmed-Head/policy-bot-desktop, mern-policy-app

FastAPI, LangChain, LLaMA Phi-3, FAISS, HuggingFace embeddings, SQL Server, MongoDB, React, .NET

- Team of four, supervised by Mr. Hassan Ali. My part: the policy data, the SQL Server schema and the RAG pipeline (embeddings, FAISS retrieval, Phi-3 running locally via llama.cpp); I also ran task planning and the report. Teammates built the Windows client, dashboards and security.
- HR policy assistant with role-based answers and audit logging; the LLM runs locally; after the FYP the team added a React / Express / MongoDB web client on the same local-LLaMA backend.

OpenClaw - Autonomous AI Publishing Assistant

Paid client delivery, March 2026

Anthropic Claude API, Node.js, Docker, Telegram Bot API, OpenAI Whisper, Hetzner VPS, Tailscale

- 24/7 assistant for a US publisher: email read/reply, voice-note-to-draft, catalogue intelligence over the full guide catalogue, royalty monitoring across multiple platforms, daily briefings. Hardened with SSH key-only auth, UFW, Fail2ban and Tailscale.

AQI Forecasting System

2025 | github.com/Danish-Ahmed-Head/aqi-predictor

Python, scikit-learn, Hopsworks, Streamlit, SHAP, GitHub Actions

- Serverless pipeline forecasting Karachi AQI 72 hours ahead; hourly feature updates, daily retraining, Ridge R-squared 0.741, live Streamlit dashboard.

Healthcare Translation Web App

2025 | github.com/Danish-Ahmed-Head/healthcare-translator

Web Speech API, JavaScript, OpenAI API (optional), Vercel

- Real-time voice-to-voice medical translation (Urdu, English, Hindi, Arabic) with dual-language transcripts, text-to-speech and multi-API fallback; live on Vercel. A working prototype, not a clinically validated device.

WhatsApp Real-Estate Lead-Qualification Agent

2026 | github.com/Danish-Ahmed-Head/whatsapp-realestate-agent

Python, FastAPI, Meta WhatsApp Cloud API

- Deterministic state-machine agent that runs keyless from the CLI, with Dubai (AED) and Karachi (PKR) locale modes.

Additional: MiniRAG: ~150-line pure-standard-library RAG demo (TF-IDF, cosine retrieval, cited answers). AI Tea Recommendation Chatbot (n8n, OpenAI, Google Sheets). Hand-coded portfolio site on GitHub Pages.

EDUCATION

Bachelor of Science in Computer Science - Karachi Institute of Economics and Technology (KIET)

Karachi, Pakistan | September 2021 - coursework completed Fall 2025

- Final Year Project: Blink (among the top five final-year projects at KIET, certificate of achievement).
- Coursework: Artificial Intelligence, Generative AI, Probability and Statistics, Data Structures and Algorithms, Database Systems, Operating Systems, Computer Networks, Parallel and Distributed Computing, Software Engineering, Cryptography.

CERTIFICATIONS

- Google Data Analytics Professional Certificate - Coursera, September 2024
- Large Language Model Operations (LLMOps) Specialization (Duke University) - Coursera, August 2024
- 31 Coursera course certificates in total (2023-2025; 14 of them are the courses of the two credentials above) from Google, Duke, IBM, Microsoft, Meta, Johns Hopkins, DeepLearning.AI and Macquarie, including: Machine Learning with Python (IBM), Databases and SQL for Data Science with Python - with honors (IBM), Data Analysis with Python (IBM), Harnessing the Power of Data with Power BI (Microsoft), AI for Everyone (DeepLearning.AI), Advanced Data Engineering and GenAI/LLMs on AWS (Duke). Credential IDs available on request.

ACHIEVEMENTS

- Among the top five final-year projects at KIET (Blink), with certificate of achievement.
- Delivered 27 scoped features in three days against a two-week client timeline (OpenClaw).
- Found and fixed 31 verified security and correctness defects in a self-directed audit of a production e-commerce codebase.
- Delivered production software for clients in the United States, Latvia and Pakistan.

LANGUAGES

English - professional working proficiency | Urdu - native | German - elementary